$$P(D \mid +c, +f, +g, +h) = \alpha P(D, +e, +f, +g, +h)$$

$$= \alpha \sum_A \sum_B \sum_C P(D, A, B, C, +e, +f, +g, +h)$$

Now factorize

$$= \alpha_D \sum_A \sum_B \sum_C P(+c \mid A, B) P(+f \mid B, C, D) P(+g \mid C, D) P(+h \mid D, +e)$$

cont → $P(A) P(B) P(C) P(D \mid A, B, C)$

Simplify

$$= \alpha_D P(+h \mid D, +e) \sum_C P(C) P(+g \mid C, D) \sum_B P(B) P(+f \mid B, C, D)$$

cont → $\sum_A P(A) P(+e \mid A, B) P(D \mid A, B, C)$

Vector form

$$= \alpha_D \langle P(+h \mid +d, +e) P(+h \mid -d, +e) \rangle \sum_C P(C) \langle P(+g \mid C, +d) P(+g \mid C, -d) \rangle$$

cont → $\sum_B P(B) \langle P(+f \mid B, C, +d) P(+f \mid B, C, -d) \rangle$

cont → $\sum_A P(A) P(+c \mid A, B) \langle P(+d \mid A, B, C) P(-d \mid A, B, C) \rangle$

Tree

$$= \alpha_D \langle P(+h \mid +d, +e) P(+h \mid -d, +e) \rangle \sum_C$$

Branch 1: $C = +c$ → $P(+c) \langle P(+g \mid +c, +d) P(+g \mid +c, -d) \rangle$

Branch 2: $C = -c$ → $P(-c) \langle P(+g \mid -c, +d) P(+g \mid -c, -d) \rangle$

Branch 1 continued (C = +c)

Branch 3

B = +b → P(+b) < P(+f | +b, +c, +d) P(+f | +b, +c, -d) >

Branch 4

B = -b → P(-b) < P(+f | -b, +c, +d) P(+f | -b, +c, -d) >

Branch 2 continued (C = -c)

Branch 5

B = +b → P(+b) < P(+f | +b, -c, +d) P(+f | +b, -c, -d) >

Branch 6

B = -b → P(-b) < P(+f | -b, -c, +d) P(+f | -b, -c, -d) >

Branch 3 continued (C = +c, B = +b)

Branch 7 A = +a → P(+a) P(+c, +a, +b) < P(+d | +a, +b, +c) P(-d | +a, +b, +c) >

Branch 8 A = -a → P(-a) P(+c, -a, +b) < P(+d | -a, +b, +c) P(-d | -a, +b, +c) >

Branch 7 = 0.11 (0.22) < 0.14, 0.86 > = < 0.003388, 0.020812 >

Branch 8 = 0.89 (0.24) < 0.18, 0.82 > = < 0.038448, 0.175152 >

Combined = < 0.041836, 0.195964 >

Branch 4 continued (C = +c, B = -b)

Branch 9 A = +a → P(+a) P(+c, +a, -b) < P(+d | +a, -b, +c) P(-d | +a, -b, +c) >

Branch 10 A = -a → P(-a) P(+c, -a, -b) < P(+d | -a, -b, +c) P(-d | -a, -b, +c) >

Branch 9 = 0.11 (0.23) < 0.16, 0.84 > = < 0.004048, 0.021252 >

Branch 10 = 0.89 (0.25) < 0.20, 0.80 > = < 0.044500, 0.178000 >

Combined = < 0.048548, 0.199252 >

Branch 5 continued (C = -C, B = +b)

Branch 11 A = +a → $P(+a)\,P(+c,+a,+b)\,\langle P(+d\mid +a,+b,-c)\,P(-d\mid +a,+b,-c)\rangle$

A

Branch 12 A = -a → $P(-a)\,P(+c,-a,+b)\,\langle P(+d\mid -a,+b,-c)\,P(-d\mid -a,+b,-c)\rangle$

Branch 11 = 0.11 (0.22) ⟨0.15, 0.85⟩ = ⟨0.003630, 0.020570⟩

Branch 12 = 0.89 (0.24) ⟨0.19, 0.81⟩ = ⟨0.040684, 0.173016⟩

Combined = ⟨0.044214, 0.193586⟩

Branch 6 continued (C = -C, B = -b)

Branch 13 A = +a → $P(+a)\,P(+c,+a,-b)\,\langle P(+d\mid +a,-b,-c)\,P(-d\mid +a,-b,-c)\rangle$

A

Branch 14 A = -a → $P(-a)\,P(+c,-a,-b)\,\langle P(+d\mid -a,-b,-c)\,P(-d\mid -a,-b,-c)\rangle$

Branch 13 = 0.11 (0.23) ⟨0.17, 0.83⟩ = ⟨0.004301, 0.020999⟩

Branch 14 = 0.89 (0.25) ⟨0.21, 0.79⟩ = ⟨0.046725, 0.175775⟩

Combined = ⟨0.051026, 0.196774⟩

Now we have summed over A so do Branch 3-6

Branch 3 = 0.12 ⟨0.26, 0.27⟩ ⟨0.041836, 0.195464⟩

= ⟨0.001305 0832, 0.006349233 6⟩

Branch 4 = 0.88 ⟨0.30, 0.31⟩ ⟨0.048548, 0.199252⟩

= ⟨0.012816672, 0.054355 9456⟩

Combined for when C = +C

= ⟨0.014 121 9552, 0.060705 1742⟩

Branch 5 = 0.12 ⟨0.28, 0.29⟩ ⟨0.044214, 0.193586⟩

= ⟨0.0014855904, 0.006736 7928⟩

Branch 6 = 0.88 ⟨0.32, 0.33⟩ ⟨0.051026, 0.196774⟩

= ⟨0.014 368 9216, 0.057 143 1496⟩

Combined for when C = -C

= ⟨0.015854512, 0.063879 4624⟩

Now we have summed over A,B so do Branch 1,2

Branch 1 = $0.13 \langle 0.34, 0.35 \rangle \langle 0.014121\,9552, 0.060705\,1742 \rangle$

$= \langle 0.000624\,19041984, 0.0027\,62085\,6536 \rangle$

Branch 2 = $0.87 \langle 0.36, 0.37 \rangle \langle 0.015854512, 0.063879\,4624 \rangle$

$= \langle 0.004965\,6331584, 0.020562\,95989\,656 \rangle$

Combined

$= \langle 0.005584\,8235\,7824, 0.023325\,0455\,5016 \rangle$

Finally, h variable

$= \langle 0.38, 0.40 \rangle \langle 0.005584\,8235\,7824, 0.023\,328\,9855\,5046 \rangle$

$P(D|+e,+f,+g,+h) = \alpha_D \langle 0.002124\,1329\,597312, 0.009\,330\,018\,220064 \rangle$

Then normalize so

D = +d

$$= \frac{0.002124\,1329\,597312}{0.002124\,1329\,597312 + 0.009\,330\,018\,220064}$$

$= 0.185446\,562\,2453203$

D = -d

$$= \frac{0.009\,330\,018\,220064}{0.002124\,1329\,597312 + 0.009\,330\,018\,220064}$$

$= 0.8145534\,4$

So, $\vec{V}$

$P(D|+e,+f,+g,+h) = \langle 0.1854465\,6, 0.8145534\,4 \rangle$

And for $D = +d = 0.185446\,562\,2453203$